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Paleomagnetism of Jurassic rocks from southwestern Sichuan and the timing of the closure of the Qinling Suture

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ABSTRACT

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A paleomagnetic study has been carried out on the Jurassic redbeds from southwestern Sichuan province on the South China Block. Thermal demagnetization has isolated a high temperature component from 20 of 24 sites from the Upper Jurassic which passes the fold and reversal tests. The tilt-corrected mean direction, computed using principal component analysis in combination with great circle intersection techniques, is $D/I = 17.9°/45.6°$ with $\alpha_{95} = 3.7°$, corresponding to a paleopole at 74.2° N/185.7° E with $A_{95} = 3.8°$. Most of the Mid-Lower Jurassic samples studied are unusually weak in magnetization and failed to give stable directions at high temperatures. However, a tilt-corrected mean direction of $D/I = 30.3°/44.6°$ ($\alpha_{95} = 13.6°$) was determined utilizing the techniques given above from 8 sites of the Lower Jurassic, which yields a paleopole at 63.8° N/188.2° E with $A_{95} = 16.1°$. This mean direction marginally passes the fold and reversal tests. A comparison with the coeval poles for the North China Block and Eurasia indicates that during the Early Jurassic these plates had not coalesced together, but that they had sutured together by the Late Jurassic. Minor but noticeable differences still exist between the Late Jurassic poles for these plates, probably due to uncertainties associated with the age of magnetization and the apparent polar wander path for Eurasia.

Introduction

The North China (Sino-Korean) Block (NCB) and the South China (Yangtze) Block (SCB) are the two major cratons that compose the eastern part of China. They are sutured together by the E–W-trending Qinling fold belt, yet there is at present no consensus on the timing of their amalgamation into the present configuration. Estimates on the age of suturing range from Devonian to Early Jurassic primarily on geologic grounds (Klimetz, 1983; Zhang et al., 1984; Mattauer et al., 1985; Sengör, 1985; Hsu et al., 1987). Increasing paleomagnetic data, however, appear to favor a younger age for the final suturing event. For example, the Late Permian and Triassic paleomagnetic poles for both blocks are statistically different (McElhinny et al., 1981; Lin et al., 1985; Opdyke et al., 1986; Zhao and Coe, 1989; Steiner et al., 1989; Yang et al., 1991a), whereas the Cretaceous paleopoles for both are virtually indistinguishable within paleomagnetic uncertainties (Kent et al., 1986; Lee et al., 1987; Enkin et al., 1991a,b). The time of the coalescence of the NCB and SCB into their present configuration thus seems to be narrowed down by paleomagnetic constraints to the Jurassic. Therefore, Jurassic paleomagnetic data becomes critical in timing this event.

Regrettably, the paleomagnetic data from the Jurassic from the NCB and SCB are still scarce, and most of those so far published are preliminary in nature according to modern reliability criteria for paleomagnetic data (Courtillot and Besse, 1986; Zhao and Coe, 1989). The situation is further complicated by the fact that the paleomagnetic signature for some reason seems difficult to retrieve from the Middle and Lower Juras-

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sic rocks from China (Enkin, 1990; Chen et al., 1991). It is extremely desirable, therefore, to obtain reliable paleomagnetic results from the Jurassic from both blocks. As part of our endeavor to contribute to the resolution of this problem, we sampled the Jurassic–Lower Tertiary redbed sequence from the Huili area of southwestern Sichuan province on the SCB. This paper reports the Jurassic results, and the Cretaceous–Lower Tertiary results will be presented elsewhere.

Geology and sampling

The sampling area is located in the western tract of the SCB, called the Kham-Yunnan Uplift (Yang et al., 1986) (Fig. 1). The basement was consolidated in the late Proterozoic (Jinningian). Immediately overlying the basement are the Sinian terrestrial acidic eruptives, clastics, tillites, and platformal marine dolomites. The Cambrian to Lower Ordovician is characterized by a thin sequence composed of shallow-marine carbonates and clastics. Middle Ordovician to Carboniferous sediments are basically missing on the Uplift. This is in sharp contrast to the adjoining East Yunnan Depression to the east and also to the Yanyuan Depression to the west which received thick marine sediments during the Late Paleozoic. In recent years, some Chinese geologists suggested that the Kham–Yunnan Uplift experienced extension and continental rifting during the Late Paleozoic through Early Mesozoic (Hercynian–Indosinian) (Cong and Huang, 1987). The tectonic and stratigraphic evolution of the Xichang–Huili area during this period was characterized by gabbroic and syenitic intrusion along the axis of the Uplift in the Permo-Carboniferous with intensive basaltic eruption in the Late Permian (Emeishan basalt). The area was dominated by fault-controlled coal-bearing molassic sedimentation in the Late Triassic. This was succeeded by the development of broader continental redbed basins in which red clastics were deposited during the Jurassic to Early Tertiary. Flat-lying lacustrine sediments of Pliocene age disconformably rest on the folded redbeds and older rocks. The folding of the sedimentary cover is, however, believed to have taken place during the Early Tertiary (Li, 1979).

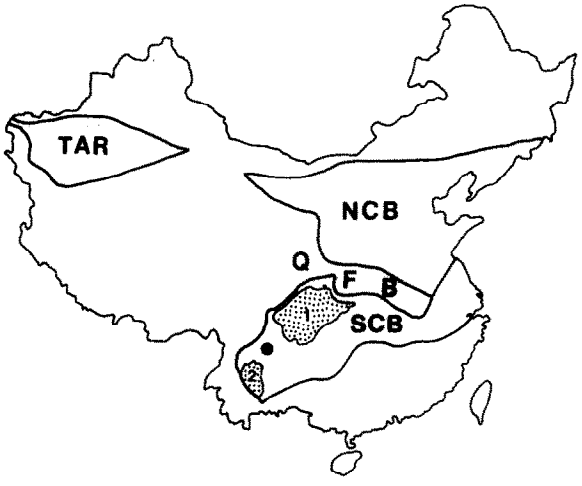


Fig. 1. Schematic map of China, showing three major Chinese cratons: *NCB* — North China Block; *SCB* — South China Block; and *TAR* — Tarim block. *QFB* denotes the Qinling fold belt separating NCB from SCB. Stippled area indicates two major Mesozoic redbed basins in western SCB: *1* = Sichuan Basin; and *2* = Central Yunnan Basin. Dot denotes the sampling area of this study.

The Xichang–Huili Jurassic–Lower Tertiary redbed sequence was deposited in a transition region between the well-known Sichuan Basin to the northeast and the Central Yunnan Basin to the southwest. The large Jurassic basin covering Sichuan and central Yunnan was probably continuous through the Xichang–Huili area (Yang et al., 1986). The palaeontologic and stratigraphic investigations conducted by the Chinese scientists in the past have established the subdivision and correlation of these redbed sequences (Sichuan Regional Stratigraphic Tables Compilation Team, 1978; Yunnan Regional Stratigraphic Tables Compilation Team, 1978; Wang et al., 1985; Hao et al., 1986). In the Huili area, the Jurassic rock formations yield many species of ostracods such as *Darwinula*, *Metacypris*, and *Limnocythere*, and bivalves such as *Pseudocardinia*, *Psilunio*, *Unio*, *Tutuella*, *Undulatula*, *Sibireconcha*, and *Sphaerium*, which commonly occur in the non-marine Jurassic sequences in South China. The Lower Jurassic Yimen Formation of the Xichang–Huili area is correlative with the Ziliujing Formation in the Sichuan Basin and ranges from 40 to 689 m in

thickness. The formation is composed of light gray and red sandstone and mudstone in the lower part, with dark red calcareous mudstone, marl and dolomite in the middle part, and the upper sequence consists of purplish red mudstone and siltstone. The Middle Jurassic Xincun Formation, 218–441 m thick, unconformably overlies the Yimen Fm. and is the stratigraphic equivalent of the Lower and Upper Shaximiao Formations plus the Xintiangou Formation in the Sichuan Basin. It is characterized by variegated feldspathic quartz sandstones and purplish red silty mudstones. The Upper Jurassic is represented by the Niugundang and Guangou Formations, which are correlative with the Suining and Penglaizhen Formations in the Sichuan Basin. The Niugundang Fm. is 180–623 m thick, composed chiefly of bright purplish red mudstone, calcareous mudstone and siltstone with intercalations of marl in the upper part. The overlying Guangou Fm. varies in thickness from 340 to 780 m, and is made up of purplish and brick red feldspathic and lithic quartz sandstone, siltstone and mudstone.

The Upper Triassic–Lower Tertiary sequence in the Huili area is exposed in an open and gentle syncline with its axis running NNE–SSW (Fig. 2). Sampling for the Jurassic was carried out at two localities. The first locality is 20 km north of downtown Huili and is called the Zhifanggou–Weijiagou ("Gou" means a ravine or gully in Chinese) section. This section is located on the western flank of the syncline, and across a fault controlled asymmetric anticline (Fig. 2). The Upper Triassic and Sinian constitute the core of the anticline with the Jurassic cropping out on the two limbs. Rocks at the west limb of the anticline dip westward at 78° to 86°, while those at the east limb dip eastward only at 27° to 36°. Ten sites (JZA-J) consisting usually of five cores 2.54 cm in diameter were drilled, using a portable rock drill and oriented with a magnetic compass,

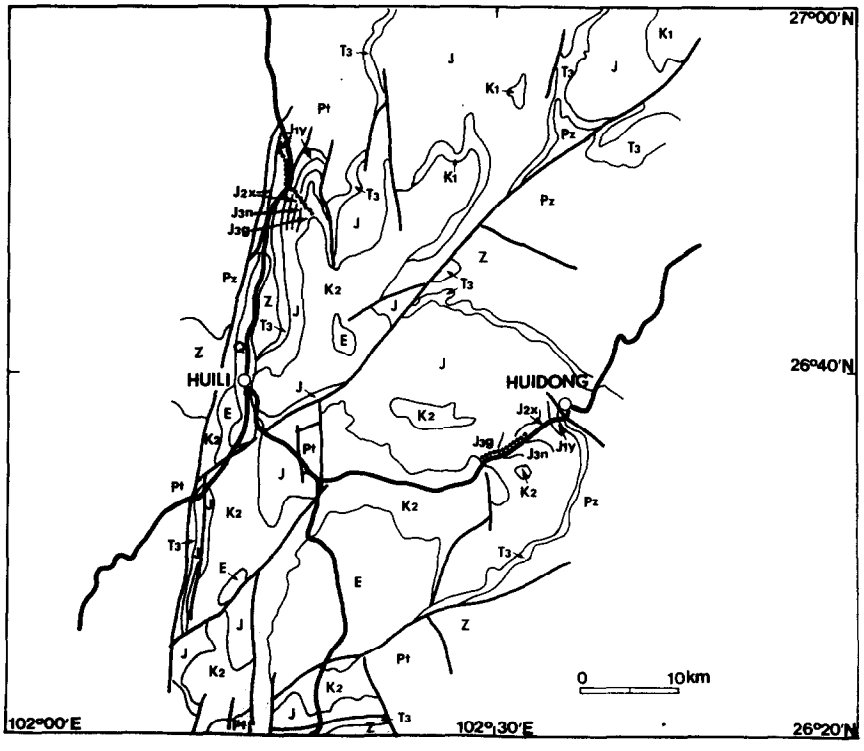


Fig. 2. Simplified geologic map of sampling area. Wide lines are major roads, and medium lines are faults. Dots denote sampling sites. Formation names: Q = Quaternary; E = Lower Tertiary; K_2 = Upper Cretaceous; K_1 = Lower Cretaceous; J = Jurassic; J_3g = Upper Jurassic Guangou Fm.; J_3n = Upper Jurassic Niugundang Fm.; J_2x = Middle Jurassic Xincun Fm.; J_1y = Lower Jurassic Yimen Fm.; T_3 = Upper Triassic; Pz = Paleozoic; Z = Sinian; and Pt = Proterozoic.

in the upper part of the Yimen Fm. on the west limb. A total of 24 sites were drilled on the east limb of the anticline, of which 5 (JZL-O) are from the Yimen Fm., 6 (JXA-F) from the Xincun Fm., and 7 each from the Niugundang (JNA-G) and Guangou (JHA-G) Formations. The second locality is 30 km east of Huili and located on the eastern flank of the regional syncline. The sampling was undertaken in the bed of a ravine along the highway from Huili to Huidong. Although the type section at which the Jurassic rock formations in the Xichang–Huili area were originally designated is 7 km farther south, the ridge-forming sandstones at the base of the Guangou and Xincun Fms. could be readily traced to the sampling site. The bedding dips southwest at 20–41°. Five sites each were collected from the Guangou (JDA-E) and Niugundang (JDF-J) Fms., and two sites (JDK-L) were from the Xincun Fm. Therefore, a fold test is possible for these sites.

Lab techniques

Lab work was carried out at the University of Florida. The samples were cut into cylinders 1.8 cm long. Thermal demagnetization was used throughout the investigation. Stepwise demagnetization was performed in a TSD-1 Schonstedt thermal demagnetizer, and the samples were usually subjected to from 7 to 19 heating steps up to 695° C. Remanence of the samples was measured in an ScT two-axis cryogenic magnetometer. All the equipment is housed in a double-walled steel room with a residual field of 50–250 nT. Demagnetization data were plotted on an orthogonal projection to examine the demagnetization behavior of the samples (Zijderveld, 1967). Paleomagnetic directions and planes were calculated using principal component analysis (PCA) (Kirschvink, 1980). Site means were computed utilizing standard Fisher (1953) statistics when only sample directions were used, or MacFadden and McElhinny's (1988) method when poles to remagnetization circles were involved. For the Upper Jurassic rocks, sample directions with maximum angular deviation (MAD) values or site mean directions with α_{95} larger than 15° were arbitrarily excluded from the calculation of site or global means. Also, samples were rejected if they had best-fit great circle paths of less than 30°.

Results

Upper Jurassic (Niugundang and Guangou Fms.)

Overall, the Upper Jurassic samples are slightly weaker in magnetization than the Cretaceous samples collected from the same area. As shown

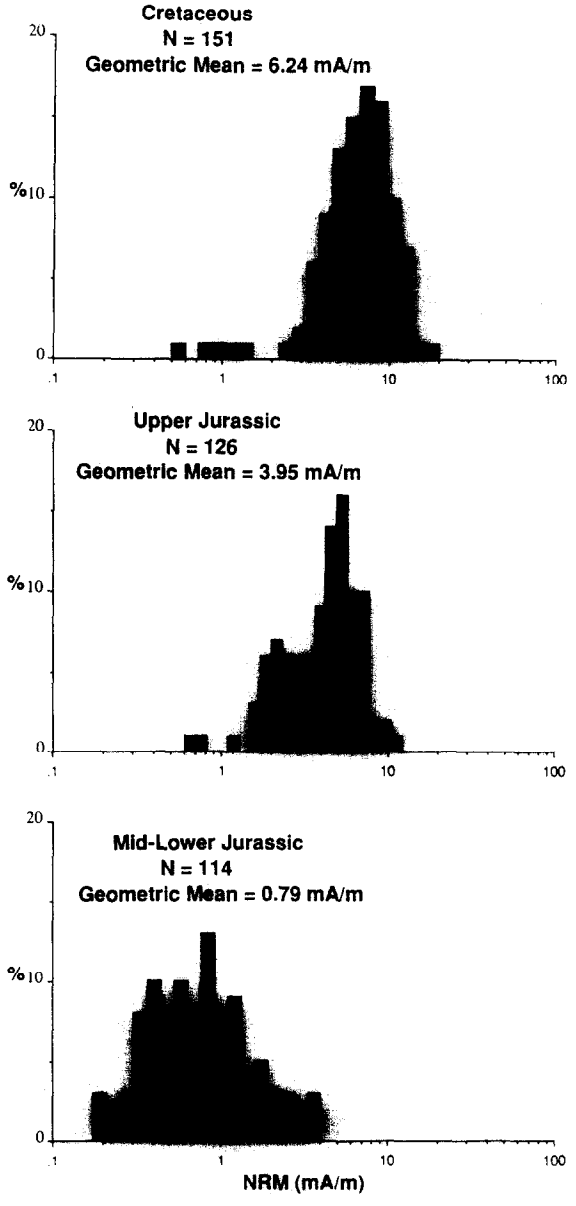


Fig. 3. Logarithmic histograms of NRM intensity of the Jurassic and Cretaceous collection from the Huili area.

in Fig. 3, the geometric mean of NRM (natural remanent magnetization) intensities for the Upper Jurassic collection is 3.95 mA/m $\pm 171\%$ (1σ) (assuming a volume of 10 cm^3 for each specimen).

Samples from 16 of the 24 sites behaved well during thermal cleaning. Most of the samples exhibited two components (Fig. 4). A lower temperature component (LTC), which generally accounts for 3–25% of the initial NRM intensity, was universally removed below 200 °C. The directions of this LTC appear to be random, and was probably acquired as a viscous remanent magnetization (VRM) during sample transportation and storage. At above 400 °C, a high-temperature component (HTC) was isolated, which decayed linearly to the origin of vector plots with increasing temperatures (Fig. 4). This component sometimes unblocked over a very small temperature range 670–690 °C (Figs. 4a, b, c). In geographic coordinates, the HTC is directed north (on the western limb of the syncline) or north-northeast (on the eastern limb) but in most cases shallower than the present dipole field direction at the sampling localities ($I = 45.2°$). After correcting for bedding tilt, all HTCs become north-northeasterly with positive inclinations.

Three other sites (JHE, JDD, and JDI) showed a second type of demagnetization behavior (Fig. 5). Samples from these sites revealed LTC and HTC similar to those of the major group described above. However, instead of linearly decaying to the origin of vector plots, the thermally distributed HTC bypassed it, and above 660 °C to 680 °C, a reversed direction emerged which could not be adequately defined with PCA technique. On the other hand, in the stereographic projection (Fig. 5), most of these samples follow well-defined great circle paths with best-fit arc length usually greater than 110°. The mean directions were determined utilizing sector-constrained great circle intersection (SCGCI) (MacFadden and McElhinny, 1988). This was possible since the sites are of reversed polarity and have small α_{95}, and are significantly different from the antipode of the present Earth field direction in geographic coordinates (Fig. 5).

Site JDH merits a detailed account. Unlike the three sites discussed above, the ChRM (characteristic remanent magnetization) component of reversed polarity was well defined in the temperature range 670–695 °C from each sample (Fig. 6). But the PCA-determined ChRM directions for the individual samples vary to a large degree, resulting in a much shallower inclination than those of the major group of normal polarity and a much larger α_{95} (25°) for the site mean direction (see Table 1). This is probably because at around the Neel temperature of hematite, the magnetic minerals become unstable. A slightly longer heating time or a small fluctuation of temperature in the heating device might trigger their sudden unblocking. As a result, if a reversed direction starts being isolated at these temperatures, the resultant direction might vary significantly between samples. Similar cases have been seen in the Cretaceous redbed samples from eastern Tibet and southern Yunnan province (our own unpublished data). On the other hand, all samples from site JDH exhibited excellent great circle paths with best-fit arc length ranging from 122° to 141° (Fig. 6). The mean direction determined using SCGCI technique is antipodal to those of the major group, and has a much smaller α_{95} (9.2°), which we believe is a better estimate of the mean direction for this site (see Table 1). This example demonstrates another advantage of MacFadden and McElhinny's method (1988).

The remaining four sites (JNC, JND, JNG and JDG) yielded large α_{95} for the site mean directions and unfortunately had to be excluded from the computation of the global mean. Samples from these sites are relatively weakly magnetized. Their paleomagnetic directions often became very erratic at higher temperatures such that either the HTCs were ill-defined or the great circle paths were very short.

Mean directions of the Upper Jurassic sites are summarized in Table 1 and are shown in equal-area projections in Fig. 7. It is clear that two polarities are present and that tilt correction substantially improves the groupings. The k_2/k_1 ratio for all the (20) sites is 3.13, higher than 2.15, the threshold for the fold test to be significant at 99% confidence level (McElhinny, 1964). Even for the major group of normal polarity, the fold

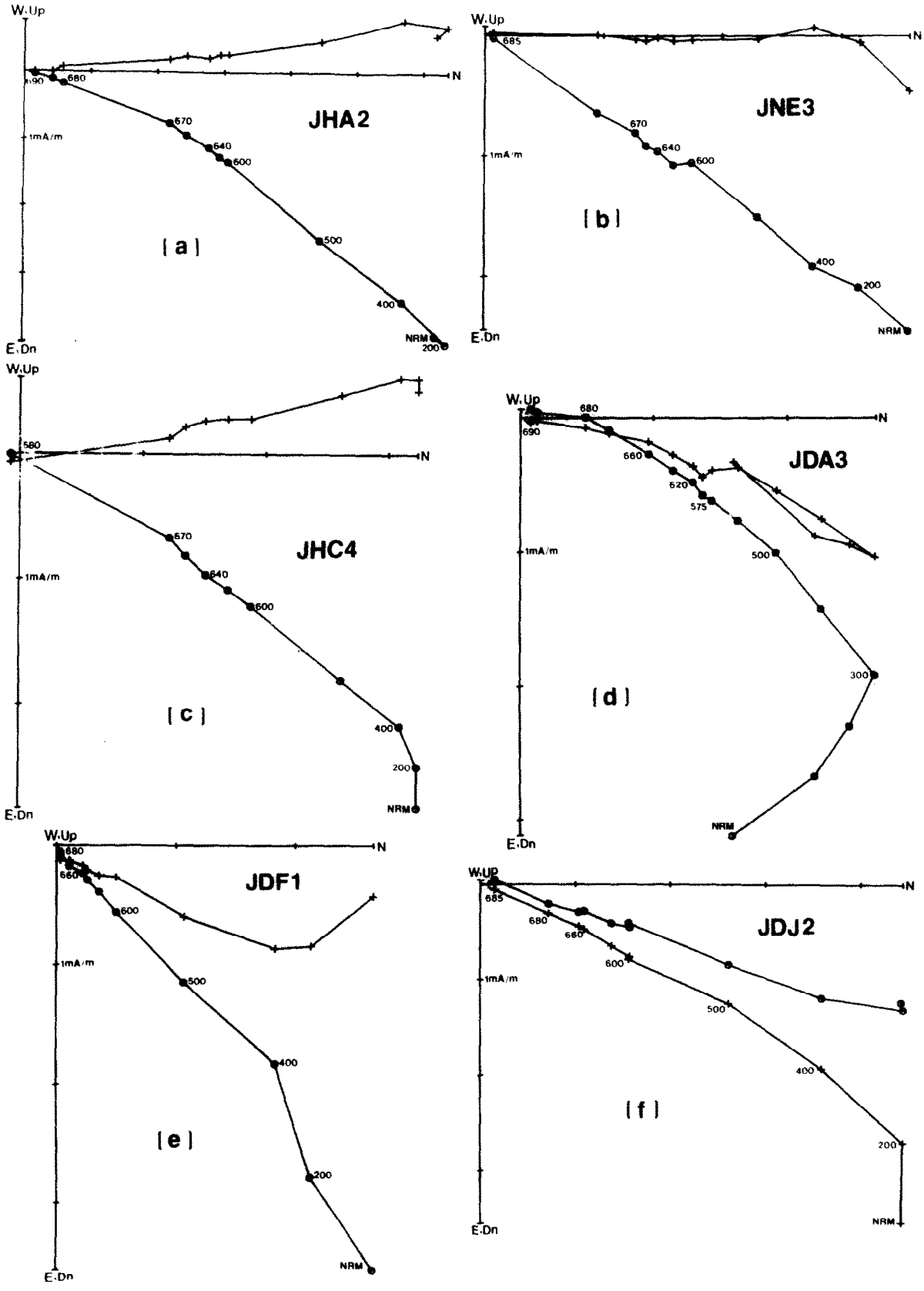


Fig. 4. Representative orthogonal plots (in geographic coordinates) for the Upper Jurassic samples. JHA2, JNE3 and JHC4 are from the western limb of the syncline, and JDA3, JDF1 and JDJ2 are from the eastern limb. Crosses (circles) are plotted on horizontal (vertical) planes. Thermal treatment levels are in degrees Celsius.

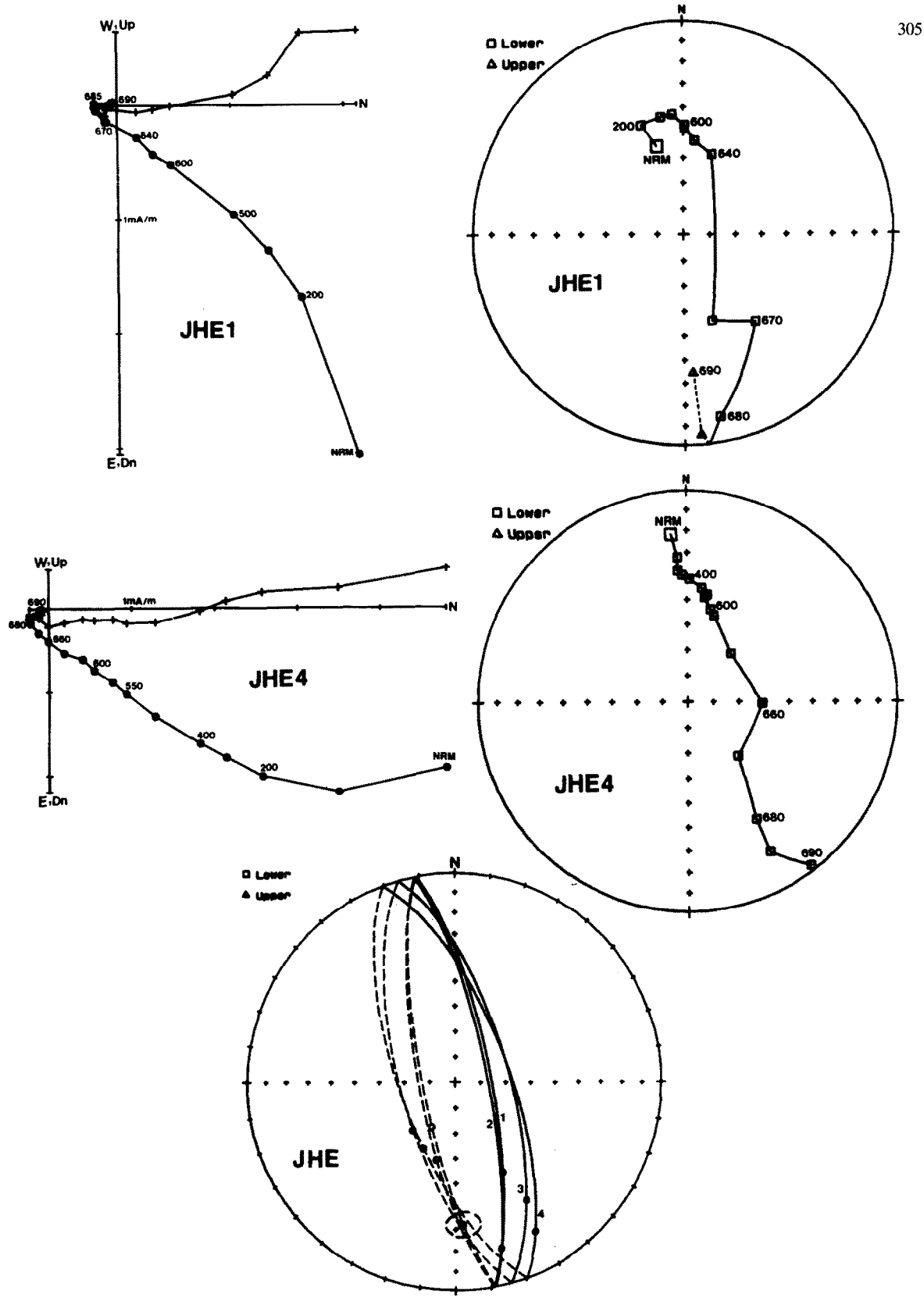


Fig. 5. Representative orthogonal and stereographic plots of sample demagnetization (upper and middle) and great circle intersection (lower) for site JHE, illustrating the demagnetization behavior of three Upper Jurassic sites whose HTCs are seen in great circle planes. All plots are in geographic coordinates. In the lower plot, sector constraints are marked with small circles on each great circle path, numbers along the path are sample numbers, and the best estimate mean for the site is indicated by the large circle (α_{95}).

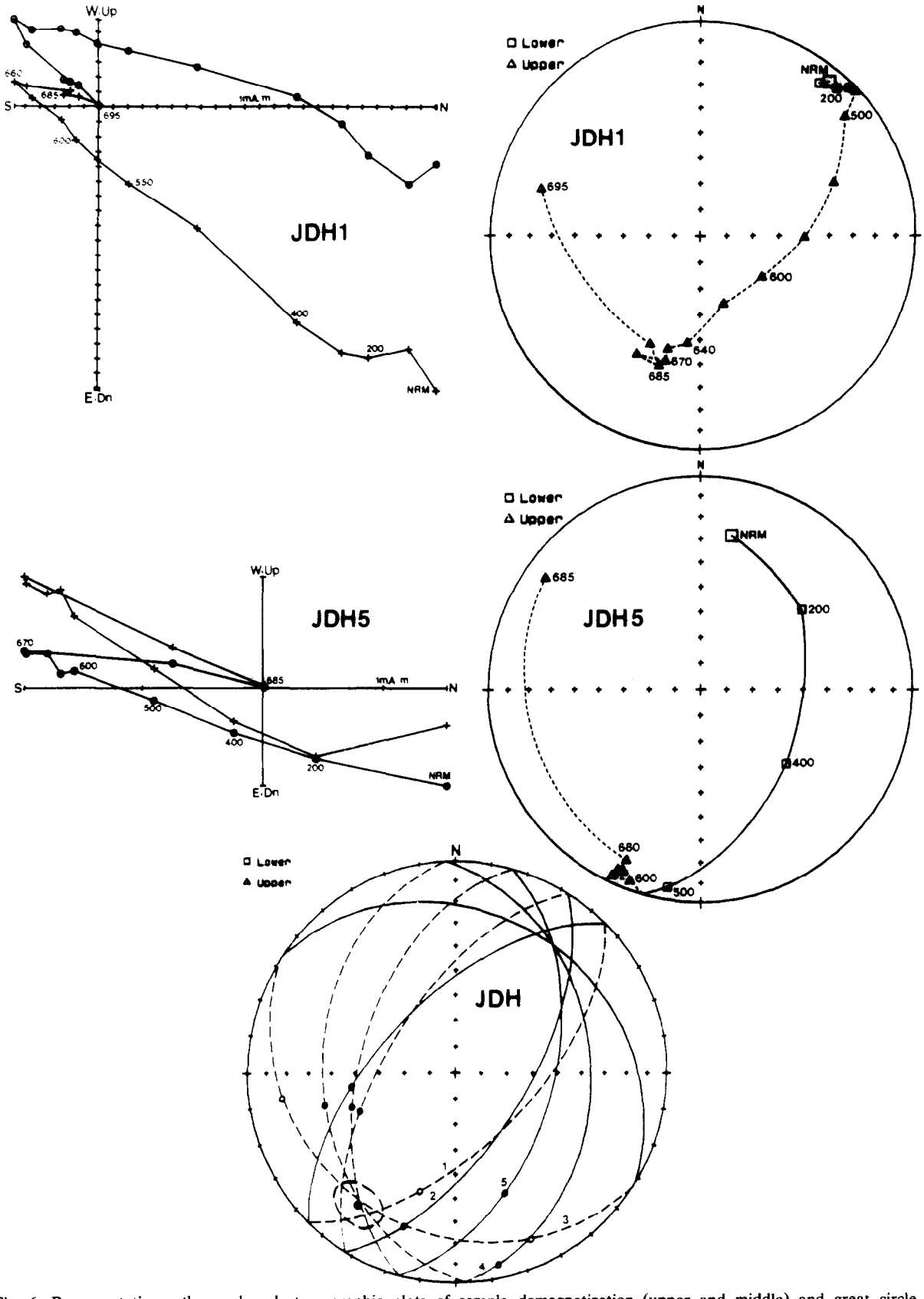


Fig. 6. Representative orthogonal and stereographic plots of sample demagnetization (upper and middle) and great circle intersection (lower) for site JDH. Coordinates and symbols as in Fig. 5. See text for further discussion.

test is also positive ($k_2/k_1 = 2.25 > 2.15$, threshold value of 99% confidence in the case of $N = 16$). Although the k_2/k_1 ratio (4.09) for the reversed group is lower than the critical value (4.28) for 95% confidence, the improvement in grouping after bedding correction is evident. Therefore, the fold test is highly positive for the Upper Jurassic formations. The tilt-corrected global mean is $D/I = 17.9°/45.6°$ with $\alpha_{95} = 3.7°$, which corresponds to a paleomagnetic pole at 74.2° N/185.7° E with $A_{95} = 3.8°$.

The angular separation between the average

TABLE 1

Statistics for the Upper Jurassic sites

Site	Bedding	n/N		In situ		Tilt corrected				Pole position	
				D (°)	I (°)	D (°)	I (°)	α_{95}	k_2	lat. (° N)	long. (° E)
Sites on the west limb of the syncline (26.8° N, 102.3° E)											
JHA	15/30	5/6		352.2	37.2	17.8	42.7	4.5	286.1	73.9	195.7
B	15/30	5/5		356.0	34.8	19.0	38.9	8.9	74.9	72.1	203.8
C	15/30	5/5		353.9	34.0	16.6	39.3	5.1	230.9	74.3	205.5
D	15/30	3/5		349.0	46.0	23.7	50.9	8.6	205.9	68.8	173.7
E	11/34	4/5	SCGCI	177.1	−32.0	200.0	−33.7	6.0	880.2	−69.8	32.5
F	11/34	5/5		3.5	36.1	27.8	33.2	6.6	137.5	63.0	205.2
G	19/31	5/5		358.5	37.3	24.6	41.5	6.1	160.9	67.7	194.4
JNA	34/35	4/5		353.2	41.8	32.4	55.7	9.5	94.0	61.0	165.4
B	20/36	6/6		348.1	36.2	20.2	46.8	6.5	106.8	72.1	183.8
C*	20/36	3/5		341.1	41.0	20.2	54.0	20.9	36.0	71.0	163.2
D*	21/31	3/5	SCGCI	171.2	−35.2	196.9	−44.8	28.4	554.5	−74.9	10.0
E	14/36	5/5		359.8	39.9	30.8	39.0	7.2	114.9	61.6	195.2
F	24/35	4/5		345.4	38.0	19.5	51.9	5.0	343.2	72.2	169.1
G*	26/34	4/6	SCGCI	155.3	−46.6	201.6	−64.2	40.9	19.7	−64.3	318.4
Sites on the east limb of the syncline (26.6° N, 102.5° E)											
JDA	110/23	5/5		5.6	25.2	5.9	43.6	7.7	99.9	84.6	203.5
B	110/23	5/5		16.3	33.0	14.5	55.9	13.1	34.9	74.3	150.5
C	110/23	6/6		10.9	29.8	7.1	52.4	4.9	189.2	81.1	144.4
D	116/20	5/5	SCGCI	188.3	−20.7	184.4	−39.8	12.7	78.0	−84.4	56.4
E	116/20	6/6		12.5	26.9	8.5	46.3	9.0	55.9	82.4	183.2
F	170/28	5/5		32.7	37.9	7.5	52.6	8.0	92.0	80.8	145.6
G*	170/28	3/5		310.6	18.9	307.0	0.2	120.5	2.2	32.6	354.0
H	156/34	5/5	SCGCI	217.0	−22.8	201.3	−50.6	9.1	150.7	−70.8	353.2
				*192.5	−12.2	180.8	−30.1	25.0	10.3	−79.5	98.3
I	156/34	6/6	SCGCI	213.9	−13.9	202.7	−41.2	9.4	87.0	−69.3	15.7
J	155/41	5/5		31.0	20.0	9.7	50.2	7.5	105.2	80.5	162.8
All sites		20		7.0	33.2	17.9	45.6	3.7		74.2	185.7
				$k_1 = 25.5$		$k_2 = 80.0$		$R_2 = 19.76$		$A_{95} = 3.8$	
Normal		16		3.5	35.5	18.1	46.6	4.1			
				$k_1 = 36.3$		$k_2 = 81.6$		$R_2 = 15.82$			
Reversed	4			199.7	−23.2	197.0	−41.6	10.8			
				$k_1 = 17.8$		$k_2 = 73.0$		$R_2 = 3.96$			

Bedding: strike/dip; n/N: number of samples or sites in calculation for the means/number of samples or sites measured; SCGCI: site mean determined from sector-constrained great circle intersection otherwise based on PCA-determined sample directions; D, I: declination and inclination; α_{95}, A_{95}: radius of the circle of 95% confidence about the mean direction and pole, respectively; k_1, k_2: Fisher precision parameters before and after tilt correction; R_2: length of resultant vector after tilt correction. *site mean excluded from calculation of the global mean.

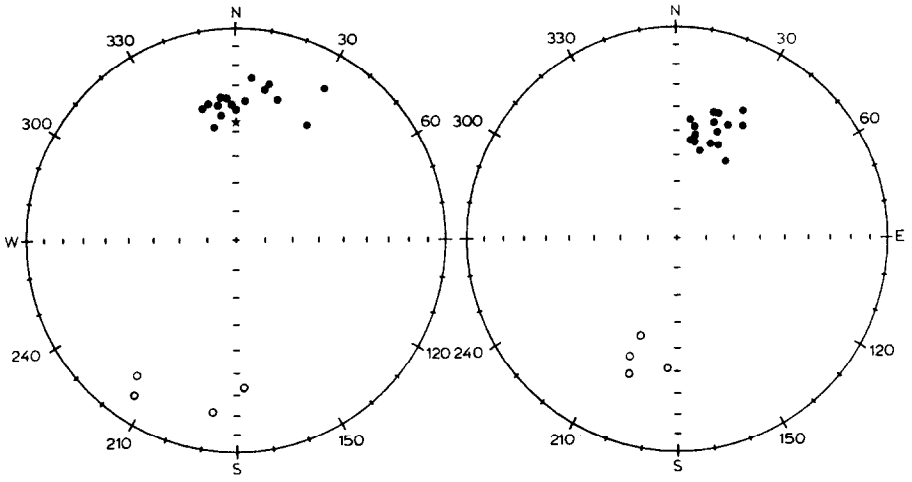


Fig. 7. Equal-area projections of site mean directions of the Upper Jurassic before (left) and after (right) tilt correction. Full (open) symbols represent directions plotted onto the lower (upper) hemisphere. Star denotes the present-day geomagnetic field direction at the sampling locality.

tilt-corrected directions of the normal and reversed groups is 5.1° after inverting one polarity, which is much smaller than the sum (14.9°) of the radii of their associated α_{95} circles, indicating that the directions of the two groups are antipodal at 95% confidence level. We also tested this in a more rigorous way. Instead of testing whether the overall mean directions from two limbs of a fold may be distinguished (MacFadden and Jones, 1981), we used the statistic $p = [R_a + R_b - R^2/(R_a + R_b)]/2(N - R_a - R_b)$, where R_a, R_b, and R are respectively the lengths of the resultant vectors of normal sites, reversed sites and all the sites (N) in this case, to check whether the

TABLE 2

Statistics for the Lower Jurassic sites

Site	Bedding	n/N		In situ		Tilt corrected				Pole position	
				D(°)	I(°)	D(°)	I(°)	α_{95}	k_2	lat.(°N)	long.(°E)
Sites on the west limb of the anticline (26.8°N, 102.3°E)											
JZA	192/77	3/3		61.6	19.8	356.6	50.7	18.1	47.5	84.5	70.3
B	200/83	4/5	SCGCI	228.8	−6.2	196.8	−29.2	25.0	50.9	−70.8	44.5
C	195/76	5/6	SCGCI	233.4	−5.2	199.5	−38.4	22.7	24.9	−71.5	24.4
D	195/76	4/5	SCGCI	245.8	26.2	241.3	−34.6	28.9	38.4	−33.5	6.4
E	189/80	5/5	SCGCI	242.4	−7.1	190.7	−53.8	14.6	59.2	−78.1	330.4
H	190/79	5/5		68.2	−27.7	62.1	40.5	11.1	48.6	34.1	181.4
Sites on the east limb of the anticline (26.8°N, 102.3°E)											
L	19/27	5/5		351.8	43.1	20.1	49.5	11.4	46.3	72.0	176.3
M	19/27	3/4		18.0	48.0	45.1	41.9	32.2	15.7	49.5	185.7
All sites		8		49.9	10.2	30.3	44.6	13.6		63.8	188.2
				$k_1 = 5.2$		$k_2 = 17.5$		$R_2 = 7.60$		$A_{95} = 16.1$	
Normal	4			39.4	25.2	33.1	48.4	23.5			
				$k_1 = 3.2$		$k_2 = 16.3$		$R_2 = 3.82$			
Reversed	4			237.4	1.8	207.8	−40.7	24.2			
				$k_1 = 20.6$		$k_2 = 15.4$		$R_2 = 3.81$			

See Table 1 for explanation.

normal and reversed directions are statistically indistinguishable. Using tilt-corrected site mean directions, $p = 0.09$ ($R_a = 15.82$, $R_b = 3.96$, $R = 19.76$, $N = 20$), less than 0.18, the 95% threshold value for significance, suggesting that they do not represent different populations of directions when inverted.

Mid–Lower Jurassic (Xincun and Yimen Fms.)

The appearance and lithology of the Mid–Lower Jurassic samples look much the same as those of the Upper Jurassic except samples from two sites (JDK and L) which are greenish in color. To our surprise, however, most of them are unusually weak in magnetization with a geometric mean of 0.79 mA/m ±200% (1σ) for the NRM intensities, almost one order of magnitude lower than that for the Upper Jurassic (Fig. 3). Moreover, a majority of these samples did not yield stable directions at higher temperatures (Figs. 8a, b).

Nonetheless, eight Lower Jurassic sites gave useable results. Samples from sites JZA, H, L and M are comparatively strongly magnetized (NRM intensities ranging from 0.68 to 3.56 mA/m) and of multicomponent nature. A soft component was eliminated below 200 °C, and a

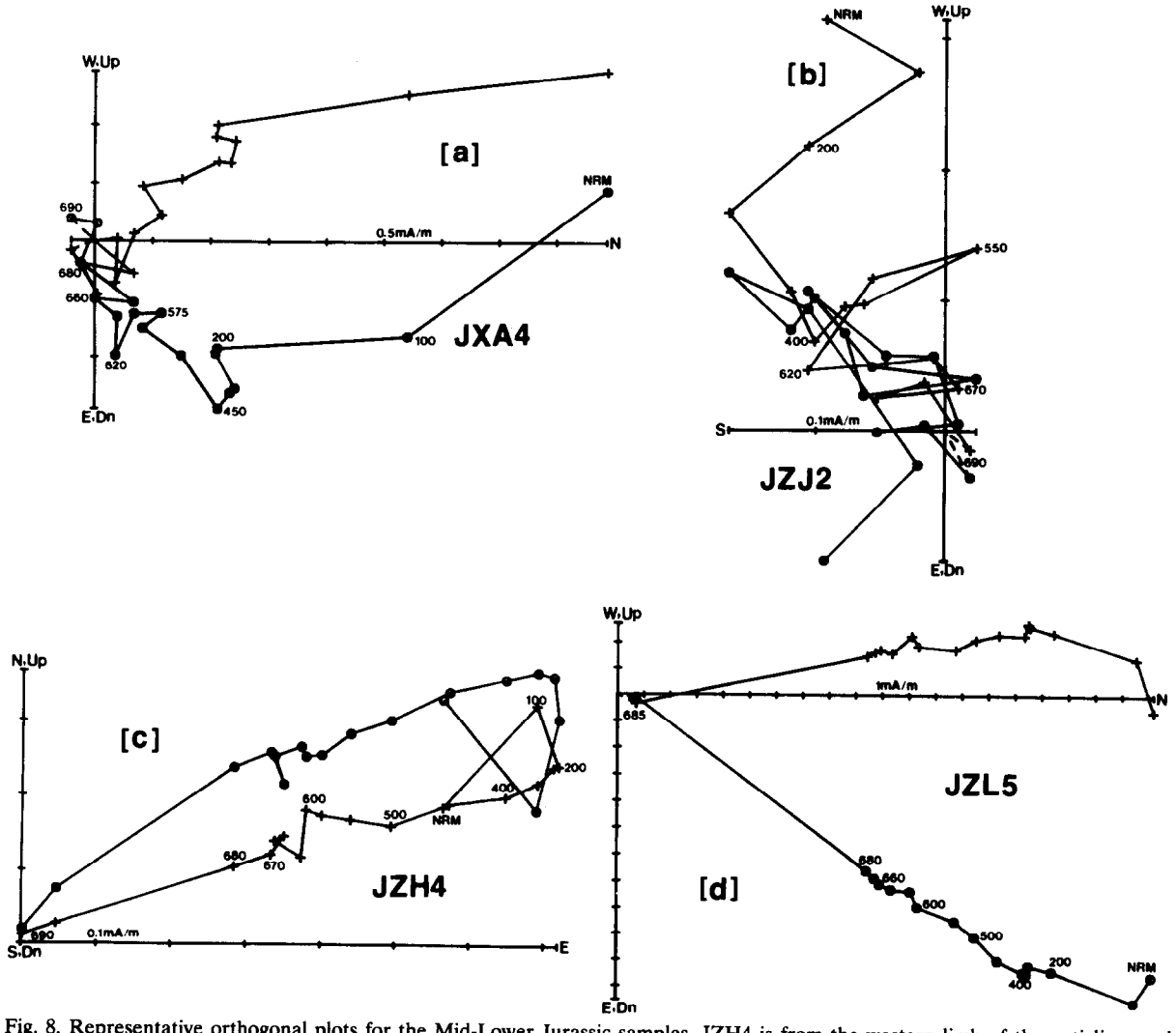


Fig. 8. Representative orthogonal plots for the Mid-Lower Jurassic samples. JZH4 is from the western limb of the anticline, and JZL5 is from the eastern limb. Coordinates and symbols as in Fig. 4.

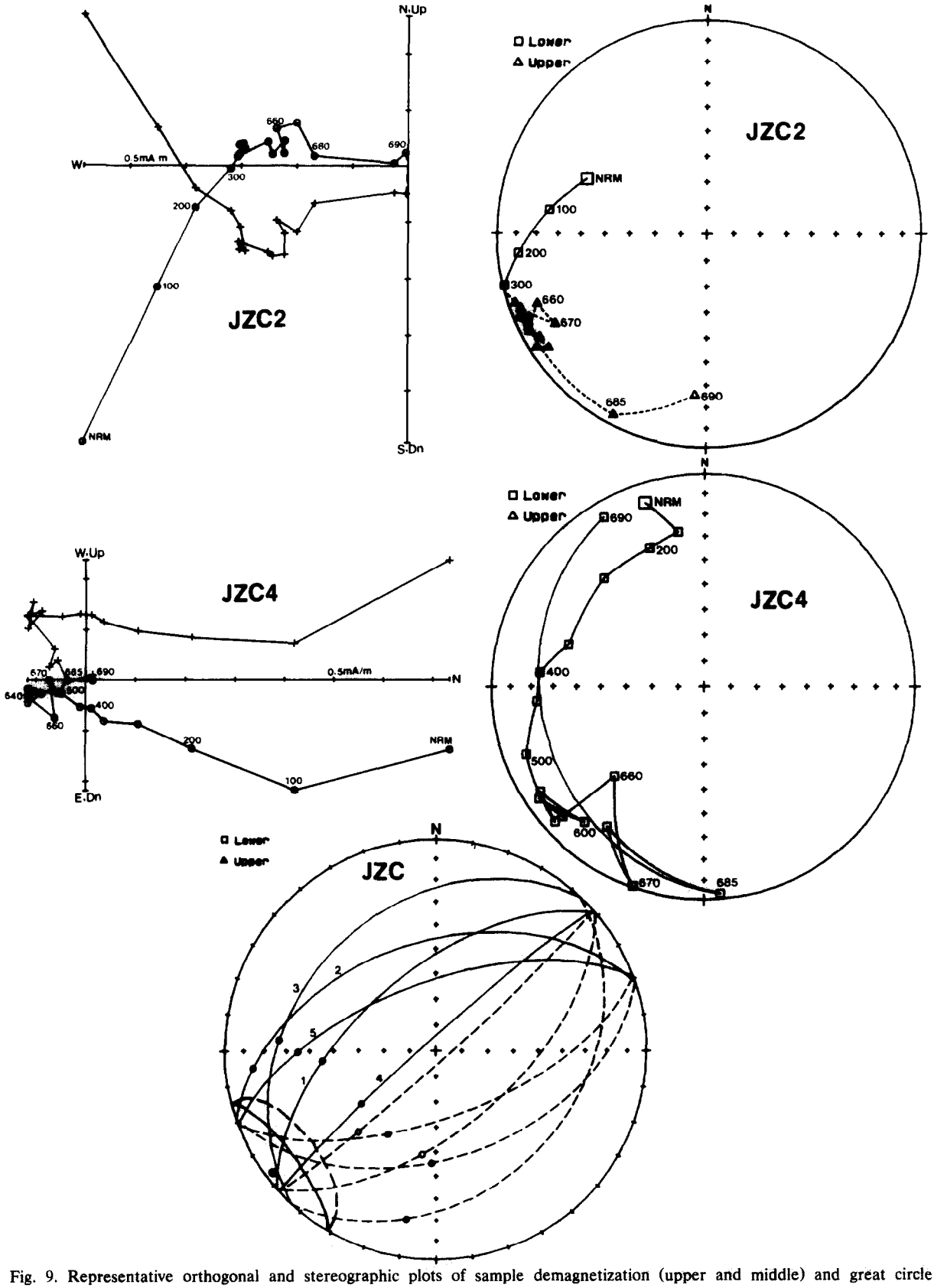


Fig. 9. Representative orthogonal and stereographic plots of sample demagnetization (upper and middle) and great circle intersection (lower) for site JZC. Coordinates and symbols as in Fig. 5.

characteristic component was isolated between 350° and 695° C (Figs. 8c, d). The in-situ ChRM directions from the two limbs of the anticline are significantly different, with those from the west limb being northeasterly and shallow up or down (Fig. 8c) and those from the east limb being northerly and moderately down (Fig. 8d). After tilt correction, they all become north- or northeast-directed with moderate positive inclinations. On the other hand, samples from sites JZB to E are rather weak in magnetization with initial intensities only in the range 0.42–1.13 mA/m. Their LTCs account for an appreciable proportion of the NRM and were apparently directed in the direction of present Earth's field in geographic coordinates (Fig. 9). At above 600° C, a second component was revealed but could not be properly determined with PCA technique, since the magnetic components became rather noisy due to weak intensities. Application of the SCGCI, with best-fit arc length mostly in the range 40–90°, yielded estimates of the hidden HTC directions for these four sites, which are southerly but very shallow in geographic coordinates and obviously not of recent origin (Fig. 9). Structural correction brings them into an antipodal group of those determined using PCA technique.

The mean directions for these eight sites are listed in Table 2 and plotted in Fig. 10. The k_2/k_1 ratio is 3.40, greater than the threshold (2.48) for significance at 95% confidence (McElhinny, 1964). Meanwhile, the calculated value for p is 0.08 ($R_a = 3.82$, $R_b = 3.81$, $R = 7.60$), lower than the critical one (0.65) for 95% confidence. Therefore, these site means pass the fold and reversal tests.

The tilt-corrected formation mean for the Lower Jurassic, based on the eight sites discussed above, is $D/I = 30.3°/44.6°$ with $\alpha_{95} = 13.6°$, corresponding to a paleopole at 63.8° N/188.2° E with $A_{95} = 16.1°$.

Discussion

The ChRM directions have been successfully determined using PCA technique from a majority of the Upper Jurassic sites studied, and utilization of MacFadden and McElhinny's method has adequately retrieved the HTC directions residing in well-defined great circle planes. The presence of two antipodal polarities indicates that enough time has lapsed during the acquisition of the magnetization to average out secular variation. A positive fold test further dates the magnetization to be prefolding, or more presicely, pre-Tertiary. Zhu et al. (1988) reported a mean direction based on five sites from the Guangou Fm. from the same area as we sampled: $D = 26.5°$, $I = 44.0°$, $\alpha_{95} = 7.6°$, which is not significantly different from ours (angular separation = $6.3 \pm 11.3°$). No

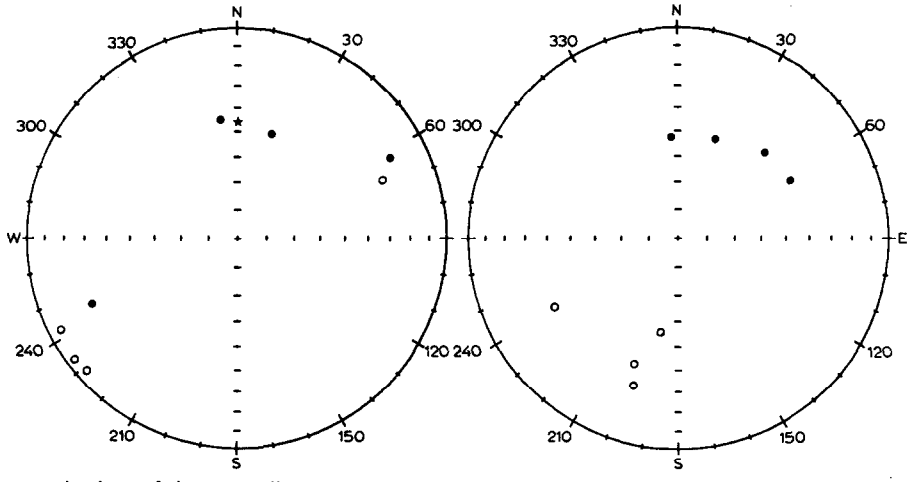


Fig. 10. Equal-area projections of site mean directions of the Lower Jurassic before (left) and after (right) tilt correction. Symbols as in Fig. 7.

details were given and no field test of stability was attempted by these workers, so their results should be superceded by ours. The angular distance between the Cretaceous pole derived from the Sichuan Basin (Enkin et al., 1991b) and the Late Jurassic pole reported here is $13.0 \pm 7.2°$. Therefore, the acquisition of the magnetization of the Upper Jurassic rocks is probably earlier than the Cretaceous.

Most of the Mid–Lower Jurassic samples we studied exhibited similar demagnetization behavior to those of the stratigraphic equivalents from the Sichuan Basin (see Enkin, 1990), and are characterized by weak magnetization and erratic direction. Nevertheless, utilization of PCA and SCGCI techniques has determined a mean direction from eight sites from the lower Jurassic, which marginally passes the fold and reversal tests. Even for the four sites from which the great circle intersection had to be resorted to retrieve the HTC directions, the unblocking temperature spectra indicate that hematite is still the magnetic carrier of the HTC like most redbeds, suggesting that the weak and noisy nature of the paleomagnetism of these rocks is probably due to lower concentration of hematite. Zhu et al. (1988) also derived a similar direction from three sites of the Lower Jurassic from the Huili–Panzhihua area: $D = 25.3°$, $I = 41.8°$, $\alpha_{95} = 10.6°$.

The available Early and Late Jurassic paleomagnetic poles for Eurasia, NCB and SCB are listed in Table 3 and plotted in Fig. 11 along with the paleopoles we presented here for comparison. It is immediately obvious that the confidence circles associated with the Early Jurassic poles for these plates do not intersect with each other despite large uncertainties (Fig. 11a), implying that these plates had not finally sutured together by the Early Jurassic. On the other hand, the Late Jurassic poles for these plates are overlapped to a first approximation (Fig. 11b), suggesting that they had docked or were closely adjacent at least by the Late Jurassic. A closer look, however, reveals some noticeable second-order differences between the Late Jurassic poles. For instance, relative to the Eurasian poles, the paleopoles for the NCB are more easterly than those for the SCB except the one derived from the Sichuan Basin (Enkin et al., 1991b). The NCB pole Zhao et al. (1990) obtained from the Inner Mongolia is a high quality one with 24 sites from 4 localities and positive fold and reversal tests. But it is significantly different from the 150-Ma pole for Eurasia (Besse and Courtillot, 1991), with angular distance between the two being $24.6 \pm 16.1°$. This difference might be due to insufficient accuracy of the Eurasian pole or to inaccurate age of the magnetization of the rocks from

TABLE 3

Paleopoles for SCB, NCB and Eurasia used in this paper

Nr.	Location	Mean pole	A_{95}	Reference
Early Jurassic				
1	SCB	63.8° N/188.2° E	16.1	This study (Table 2)
2	SCB	67.4° N/192.4° E	11.4	Recalculated from Zhu et al. (1988)
3	NCB	82.4° N/286.0° E	7.1 [a]	Yang et al. (1991b)
4	Eurasia (190 Ma)	66.7° N/109.0° E	3.9	Besse and Courtillot (1991)
Late Jurassic				
5	SCB	74.2° N/185.7° E	3.8	This study (Table 1)
6	SCB	66.3° N/187.2° E	9.4	Recalculated from Zhu et al. (1988)
7	SCB	74.6° N/235.3° E	11.5 [a]	Enkin et al. (1991b)
8	SCB	73.0° N/213.7° E	12.6	Lin et al. (1985)
9	NCB	68.5° N/231.6° E	9.5	Zhao et al. (1990)
10	NCB	71.0° N/222.2° E	8.9	Recalculated from Lin (1984)
11	Eurasia (150 Ma)	70.0° N/157.8° E	6.6	Besse and Courtillot (1991)
12	Eurasia (140 Ma)	71.6° N/173.0° E	10.4	Besse and Courtillot (1991)

Nr.: Numbers denoting paleopoles in Fig. 11. [a] estimated from original dm and dp values.

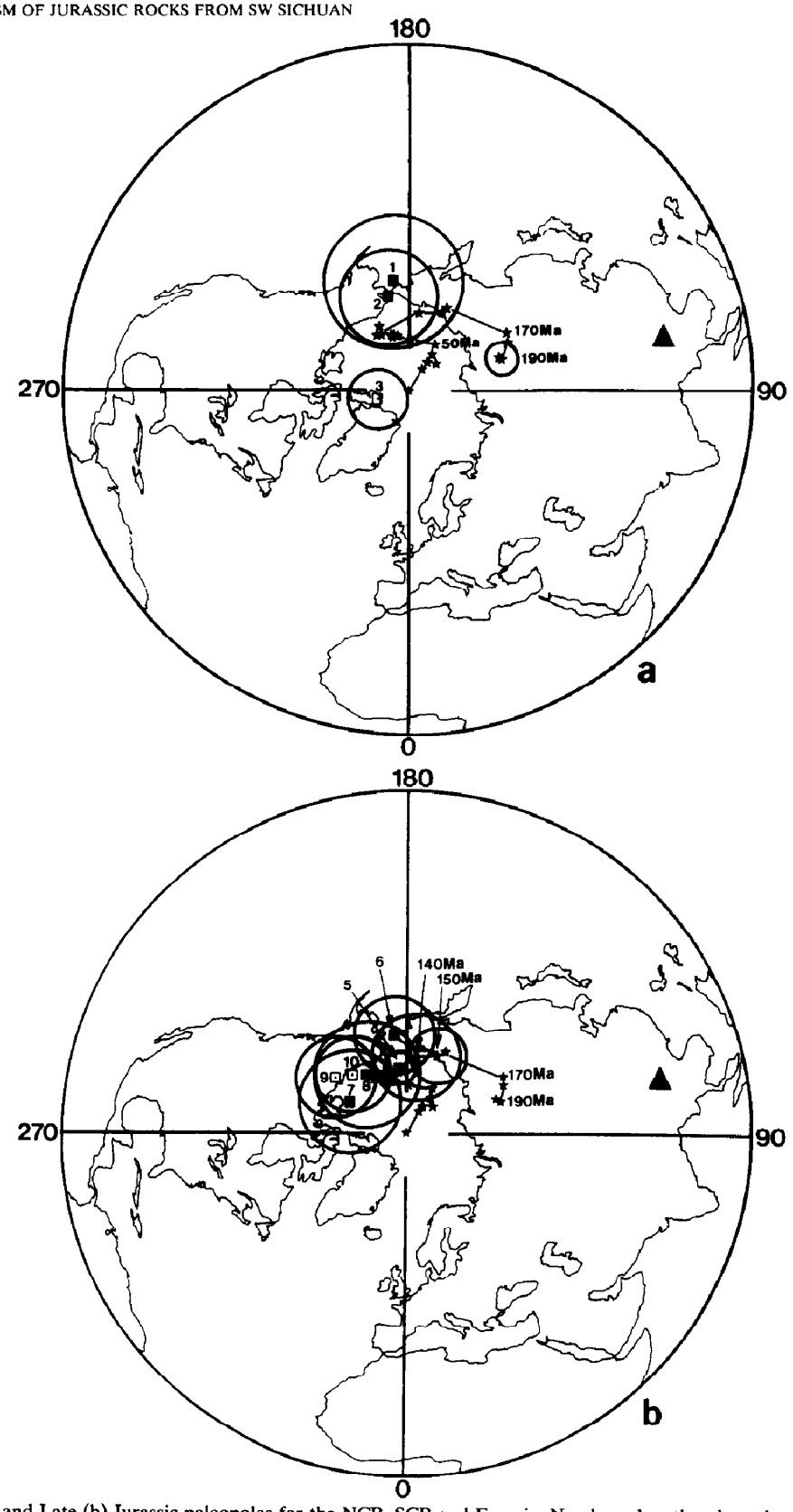


Fig. 11. Early (a) and Late (b) Jurassic paleopoles for the NCB, SCB and Eurasia. Numbers denoting the poles are the same as in Table 3. The Besse and Courtillot (1991) Eurasian APWP is marked in stars. The paleopoles for the SCB and NCB are marked with solid and open squares, respectively. The black triangle denotes the sampling locality of this study. Equal-area projection.

which Zhao et al.'s pole was determined. The synthetic apparent polar wander path (APWP) for Eurasia proposed by Besse and Courtillot (1991) is an updated one. They first select higher quality paleomagnetic poles by applying more stringent selection criteria, and then transfer the data using ocean kinematic models to a common frame (Africa), and finally average these transferred data in 20-Ma windows to generate a synthetic APWP for each plate considered. From the synthetic APWP, they discovered a previously ill-recognized hairpin turn at 80–130 Ma for Eurasia. But as they pointed out, the Jurassic poles they used for Eurasia (and Africa) are few, so the poles of this period for the synthetic APWP for Eurasia rely heavily on those transferred from North America. The locations of the transferred North American Mid-Late Jurassic poles as well as the European poles themselves, however, are generally more westerly than those from China and Korea (Kim and Van der Voo, 1990). In fact, Besse and Courtillot's 140-Ma pole has a large α_{95} confidence circle (10.4°), and is indistinguishable from Zhao et al's Late Jurassic pole for the NCB (angular distance = 19.4 ± 19.8°), probably a reflection of the uncertainties associated with the synthetic APWP for Eurasia for this particular period. We also noticed that our Late Jurassic pole for the SCB is distinct from Zhao et al.'s pole for the NCB with an angular distance of 15.3 ± 13.3° between the two. The difference is essentially in paleolatitude. The paleolatitude determined from our results for the Huili area is almost the same as today, whereas the paleolatitude calculated from Zhao et al.'s pole is 13–14° more south than the present latitude of their sampling sites. If the hairpin turn of the APWP for Eurasia is a real feature, the locations of the paleopole for this time interval determined from individual blocks which supposedly attached to Eurasia would be very sensitive to the age of the magnetization from which the paleopole is computed. Following in this vein, the Late Jurassic pole we obtained for the SCB might be representative of the Jurassic/Cretaceous boundary, whereas the pole Zhao et al. determined for the NCB might be a Cretaceous remagnetization since it is close to the tip of the hairpin turn of Besse and Courtillot's APWP for Eurasia. At present these possibilities can not be tested.

The estimation of a Paleozoic age for the final suturing between the NCB and SCB was primarily based upon structural geology and stratigraphic analyses. The granitoid intrusions of Yanshanian (Jurassic–Cretaceous) age represented by huge batholiths and numerous small stocks, however, are well developed in the middle portion of the Qinling fold belt. The Yanshanian granitic intrusions are also prominent in the eastern portion of the Fold Belt (Yang et al., 1986). The E–W-trending distribution of these intrusives cannot simply be related to the westward subduction of the Pacific (Kula) plate under the Asian continent during the Late Mesozoic. Instead, they might be products of the subduction between the SCB and NCB. Unfortunately, these granitic intrusives have not been intensively studied petrologically and geochemically. It is suggested that a study of this kind with an aim of addressing their tectonic settings may help solve this discrepancy. The paleomagnetic data now available favors a pre-Late, probably Middle, Jurassic suturing of the North and South China blocks along the Quiling suture (Lin et al., 1985; Opdyke et al., 1986; Zhao and Coe, 1989; Steiner et al., 1989; Enkin, 1990; Yang et al., 1991a,b). Small post Jurassic motion is, however, possible.

Conclusion

We have determined the paleomagnetic directions from the Jurassic redbeds from southwestern Sichuan on the SCB. The Upper Jurassic directions pass the fold and reversal tests at high confidence, whereas the Lower Jurassic directions pass these stability tests only marginally. Comparison of the paleomagnetic poles derived from these directions and the coeval poles for the NCB and Eurasia indicates that these plates had sutured together by the Late Jurassic but probably not during the Early Jurassic. Minor differences between the Late Jurassic poles for these plates may be due to the uncertainties associated with the APWP for Eurasia and the age of the magnetization from which the paleopoles for the NCB and SCB were determined. More paleomag-

netic data from the Jurassic is needed to further resolve these discrepancies.

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